Lecture-04 Date: 14.07.26

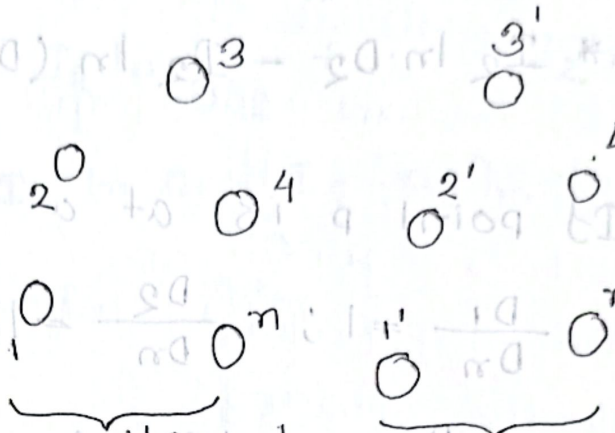
Art 2.7

## Inductance of Composite Conductor Wires

Flux linkage of one conductor in a group

$$\lambda_i = 2\times10^{-7}\left[I_1 \ln\left(\frac{1}{D_{i1}}\right) + I_2 \ln\left(\frac{1}{D_{i2}}\right) + \cdots + I_i \ln\left(\frac{1}{r_i'}\right) + \cdots + I_n \ln\left(\frac{1}{D_{in}}\right)\right]$$

Filament '1' of conductor 'A': Flux linkage




Composite Conductor A
- → 'n' identical strand or filaments
- → Radius remain Same.
- → Total current I
- → Carry current per filament.

Conductor B
- → 'm' identical strands or filament
- → Current returning -I
- → Total Current -I
- → Return current ~~for~~ per filament $\frac{-I}{m}$

$$\lambda_1 = \underline{2\times10^{-7}\frac{I}{n}\left[\ln\left(\frac{1}{r_1'}\right) + \ln\left(\frac{1}{D_{12}}\right) + \ln\left(\frac{1}{D_{13}}\right) + \cdots + \ln\left(\frac{1}{D_{1n}}\right)\right]} - \underline{2\times10^{-7}\frac{I}{m}\left[\ln\left(\frac{1}{D_{11'}}\right) + \ln\left(\frac{1}{D_{12'}}\right) + \cdots + \ln\left(\frac{1}{D_{1m}}\right)\right]}$$

Within group / ~~Between~~ two group — between two group

$$= 2\times10^{-7}\frac{I}{n}\left[\ln\left[\frac{1}{r_1' D_{12} D_{13} \cdots D_{1n}}\right]\right] - 2\times10^{-7}\frac{I}{m}\ln\left[\frac{1}{D_{11'} D_{12'} D_{13'} \cdots D_{1m}}\right]$$

$$= 2\times10^{-7} I \ln\left[\frac{1}{r_1' D_{12} D_{13} \cdots D_{1n}}\right]^{1/n} - 2\times10^{-7} I \ln\left[\frac{1}{D_{11}' D_{12}' \cdots D_{1m}}\right]^{1/m}$$

→ Mutal

$$= 2\times10^{-7} I \ln\left[\frac{(D_{11}' D_{12}' D_{13}' \cdots D_{1m})^{1/m}}{(r_1' D_{12} D_{13} \cdots D_{1n})^{1/n}}\right]$$

→ Self

$$L_1 = \frac{\lambda_1}{I/n} = 2\times10^{-7} n \ln\left[\frac{\sqrt[m]{D_{11}' D_{12}' D_{13}' \cdots D_{1m}}}{\sqrt[n]{r_1' D_{12} D_{13} \cdots D_{1n}}}\right]$$

$$L_2 = \frac{\lambda_2}{I/n} = 2\times10^{-7} n \ln\left[\frac{\sqrt[m]{D_{21}' D_{22}' D_{23}' \cdots D_{2m}}}{\sqrt[n]{r_2' D_{22} D_{23} \cdots D_{2n}}}\right]$$

Inductance of filament 'i'

$$L_i = \frac{\lambda_i}{I/n} = 2\times10^{-7} n \ln\left[\frac{\sqrt[m]{D_{i1}' D_{i2}' D_{i3}' \cdots D_{im}}}{\sqrt[n]{D_{i1} D_{i2} D_{i3} \cdots D_{in}}}\right]$$

Average inductance of the filaments of composite conductor A

$$L_{avg} = \frac{L_1 + L_2 + \cdots + L_n}{n}$$

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$$= 2\times10^{-7}\left[\frac{\sqrt[m]{(D_{11}'\cdots D_{1m})(D_{21}'\cdots D_{2m})\cdots(D_{n1}'\cdots D_{nm}')}}{\sqrt[n]{(D_{11}\cdots D_{1m})(D_{21}\cdots D_{2n})\cdots(D_{n1}\cdots D_{nm})}}\right]$$

Mutual

Self

Inductance of Conductor A

$$L_A = \frac{L_{AVG}}{n} = \frac{L_1+L_2+\cdots+L_n}{n^2}$$

$$L_A = 2\times10^{-7}\ln\left[\frac{\sqrt[mn]{(D_{11}'\cdots D_{1m})(D_{21}'\cdots D_{2m})}}{\sqrt[n^2]{(D_{11}\cdots D_{1n})(D_{21}\cdots D_{2n})}}\right]$$

$$\left[\frac{\cdots(D_{n1}'\cdots D_{nm}')}{\cdots(D_{n1}\cdots D_{nm})}\right]$$

$$= 2\times10^{-7}\ln\left(\frac{D_m}{D_s}\right)\ H/m$$

$$= 0.461\log\left(\frac{D_m}{D_s}\right)$$

$D_m$ = Mutual Geometric Mean distance

$D_s$ = Self Geometric Mean Distance

For Exam : Find $D_m$, $D_s$ type Math from Example, Exercise

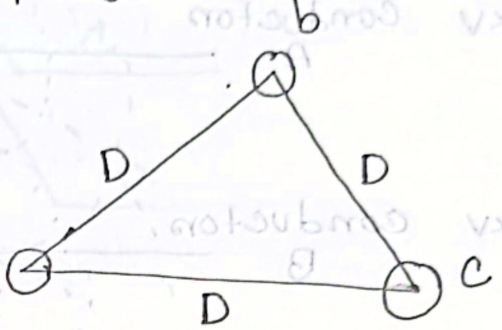
*** Derivation very important for Exam

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## Art - 2.8 : 3-$\phi$ Line with Symmetrical Spacing

Total flux of conductor phase 'a'

$$\lambda_a = 0.4605 \left[ I_a \log\left(\frac{1}{r'}\right) + I_b \log\left(\frac{1}{D}\right) + I_c \log\left(\frac{1}{D}\right) \right]$$




As, loads are balanced

Radius of conductor = r

$$I_a + I_b + I_c = 0$$

$$I_b + I_c = -I_a$$

$$\lambda_a = 0.4605 \left[ I_a \log\left(\frac{1}{r'}\right) - I_a \log\left(\frac{1}{D}\right) \right]$$

$$= 0.4605 \left[ I_a \log\left(\frac{D}{r'}\right) \right] \text{ mWb-T/km} \quad - (1)$$

$$L_a = \lambda_a / I_a = 0.4605 \log\left(\frac{D}{r'}\right) \text{ mH/km}$$

$$L_b = \frac{\lambda_b}{I_b} = 0.4605 \log\left(\frac{D}{r'}\right) \text{ mH/km}$$

$\lambda_a = \lambda_b = \lambda_c$

$L_a = L_b = L_c$

Inductances are Identical.

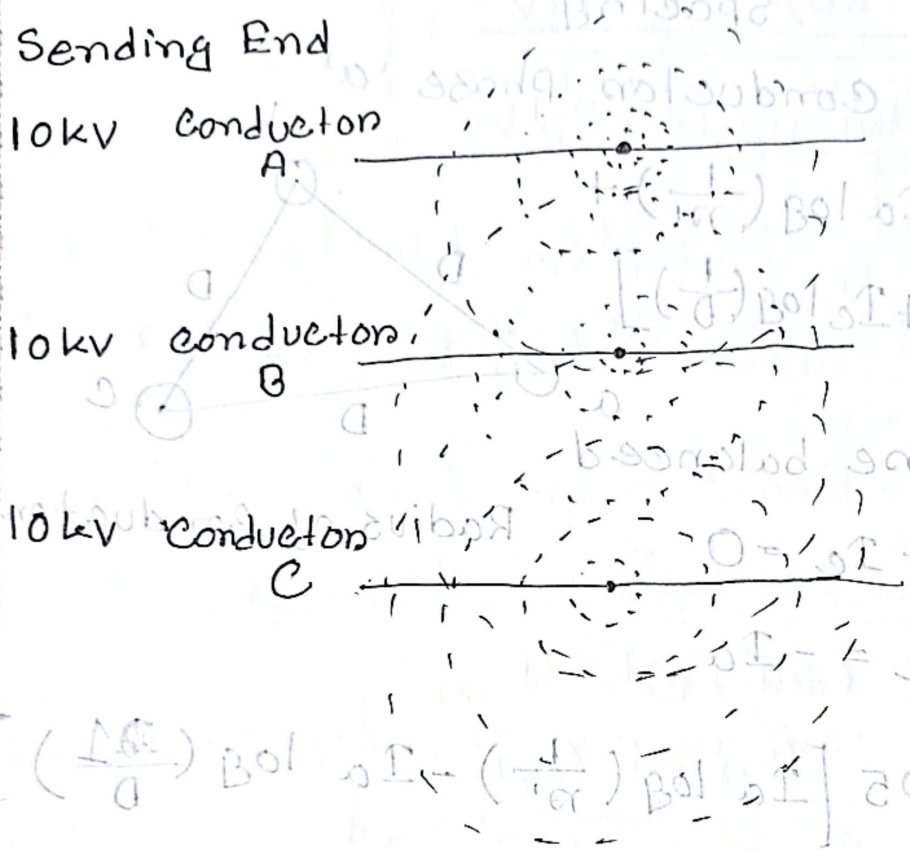
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Art-2.8

## Transpose transmission Line

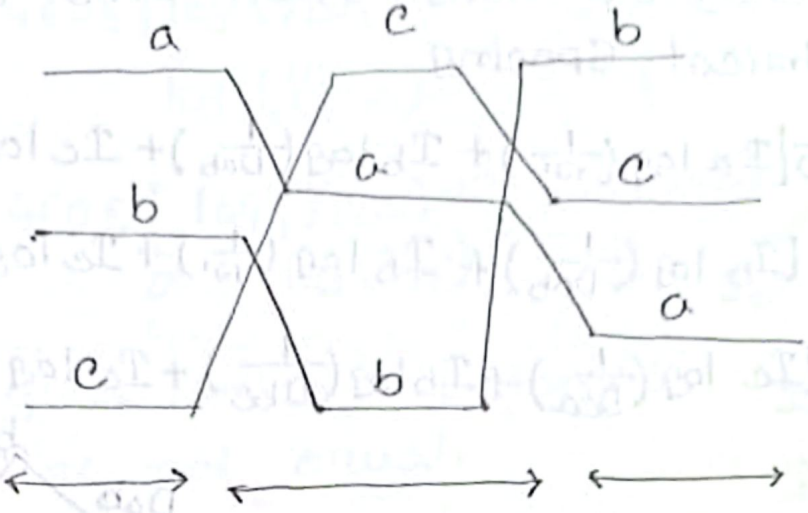
Sending End

Receving End




Observations :

- At middle conducton : Flux linkage high
- Due to large flux linkage resistance will be greater than other two conducton
- Large power loss




Section-1 Section-2 Sec-3

a-b-c c-a-b b-ca

Transpose Transmission Line :-